

Image Formation

The Eye and The Camera

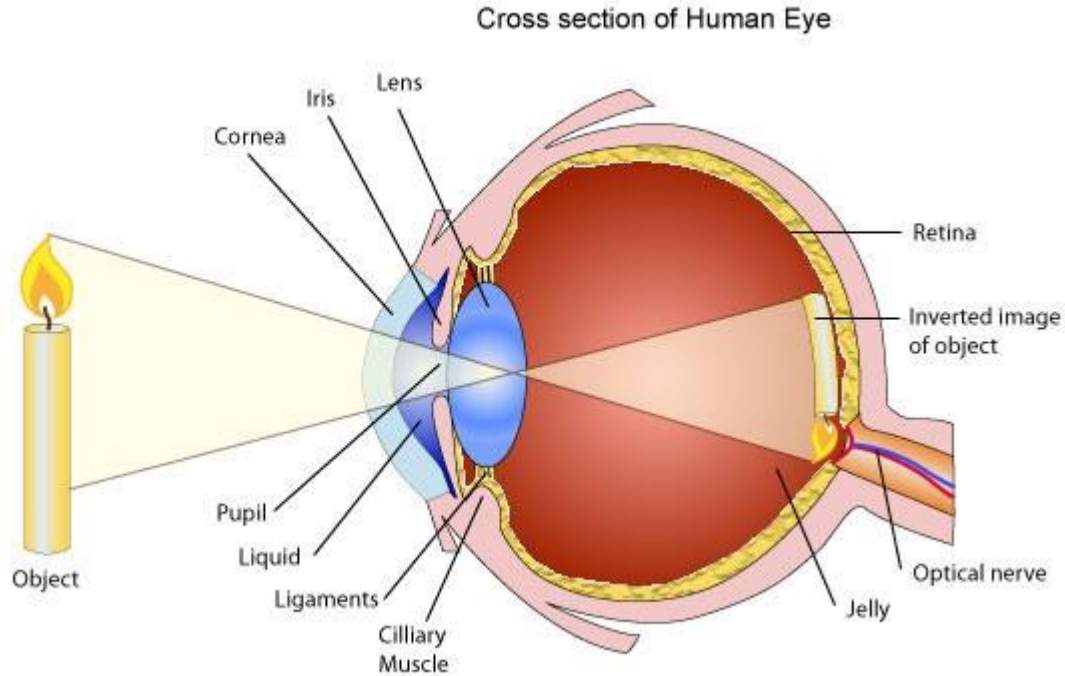
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The University of Sheffield

Outline

- Human Vision
- Camera
- Image Formation
 - Resolution
 - Bit-depth
- Color
 - Grayscale – color

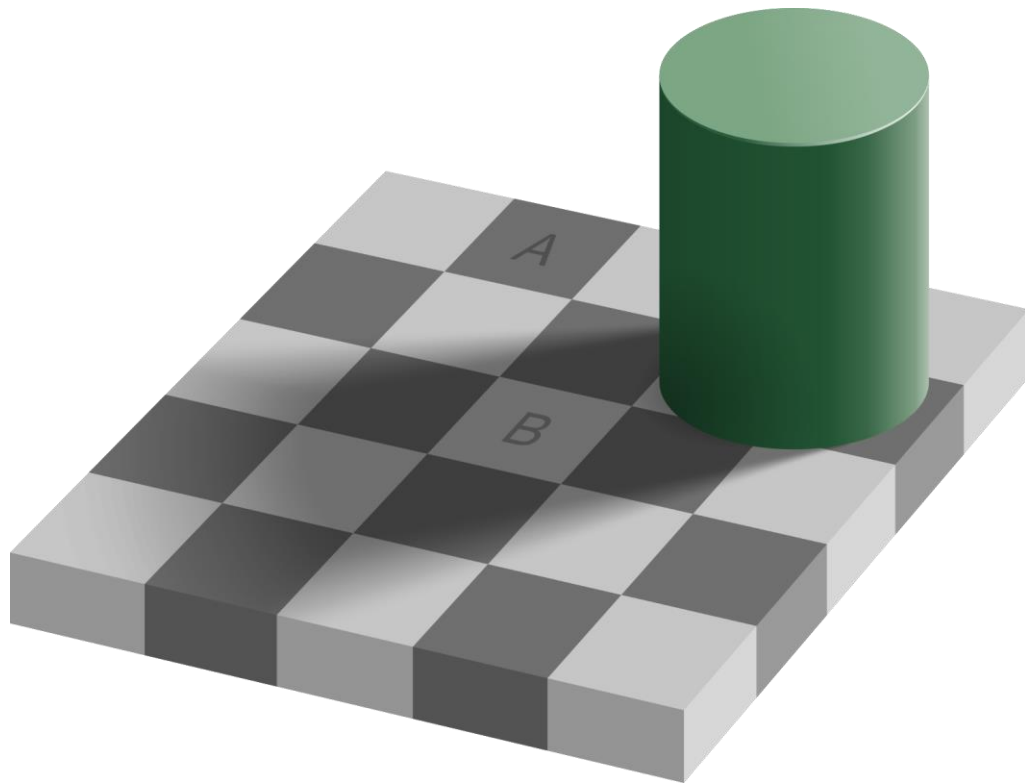


A Sensor: Human Eye

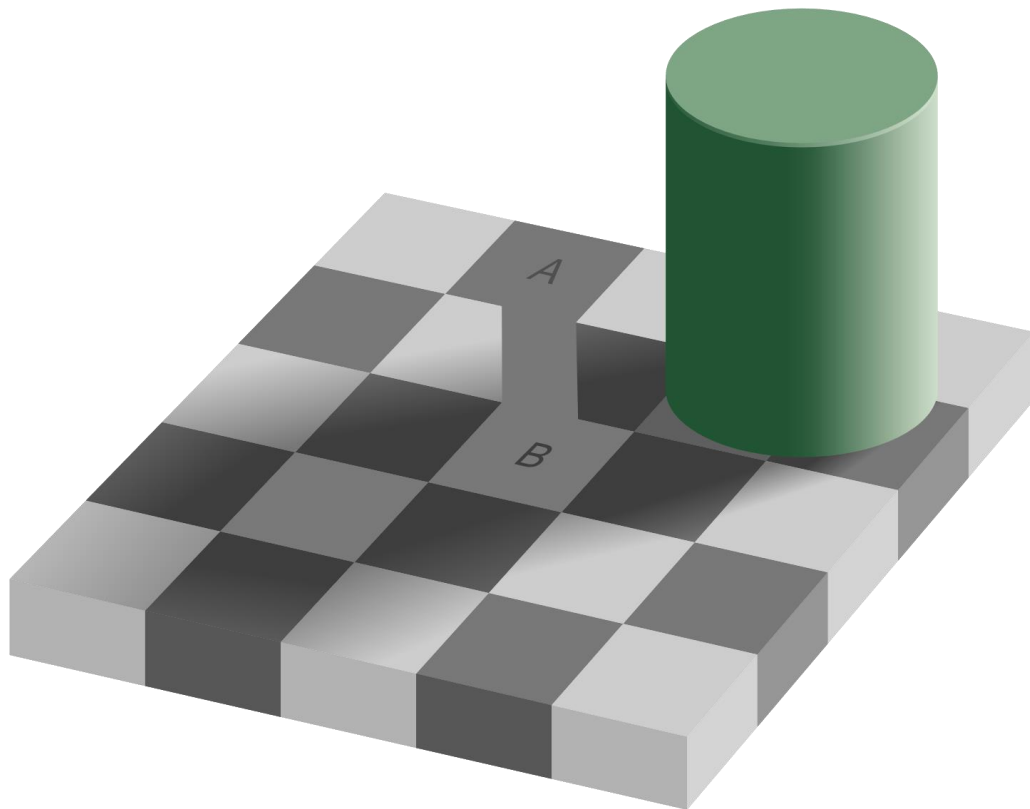


Source: <http://rmsphotographygcse.weebly.com>

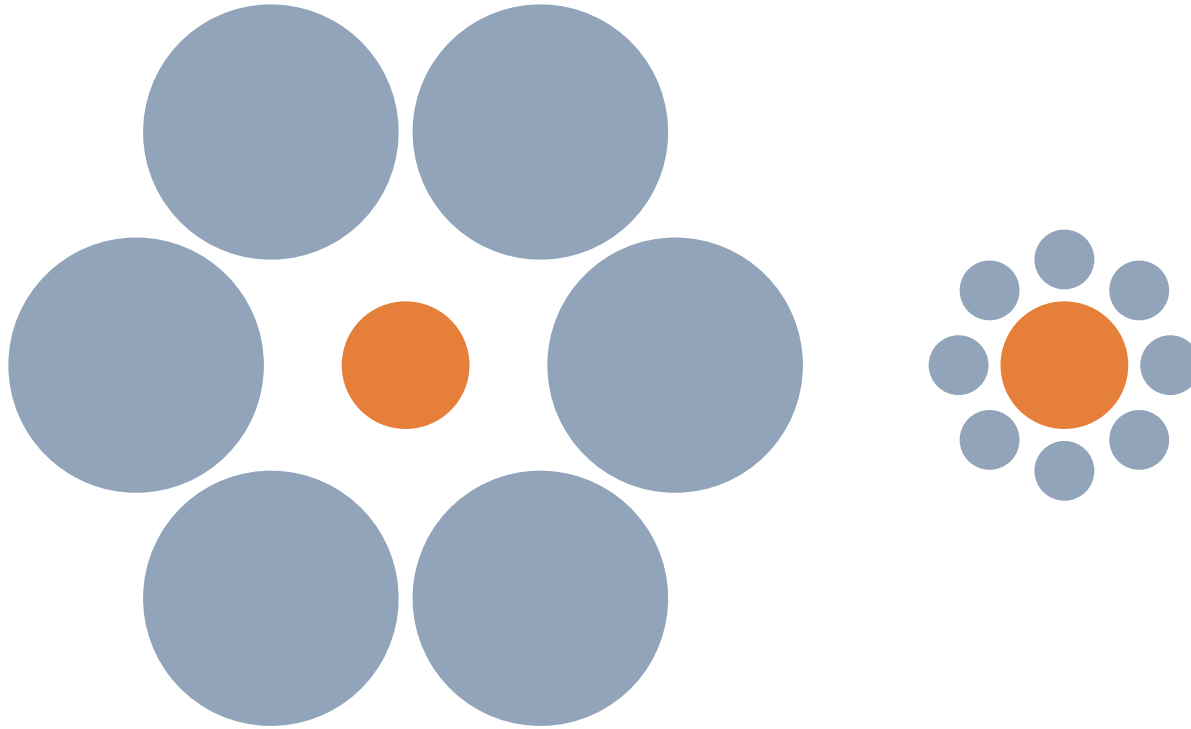
Brighter, A or B? (Checker shadow illusion)



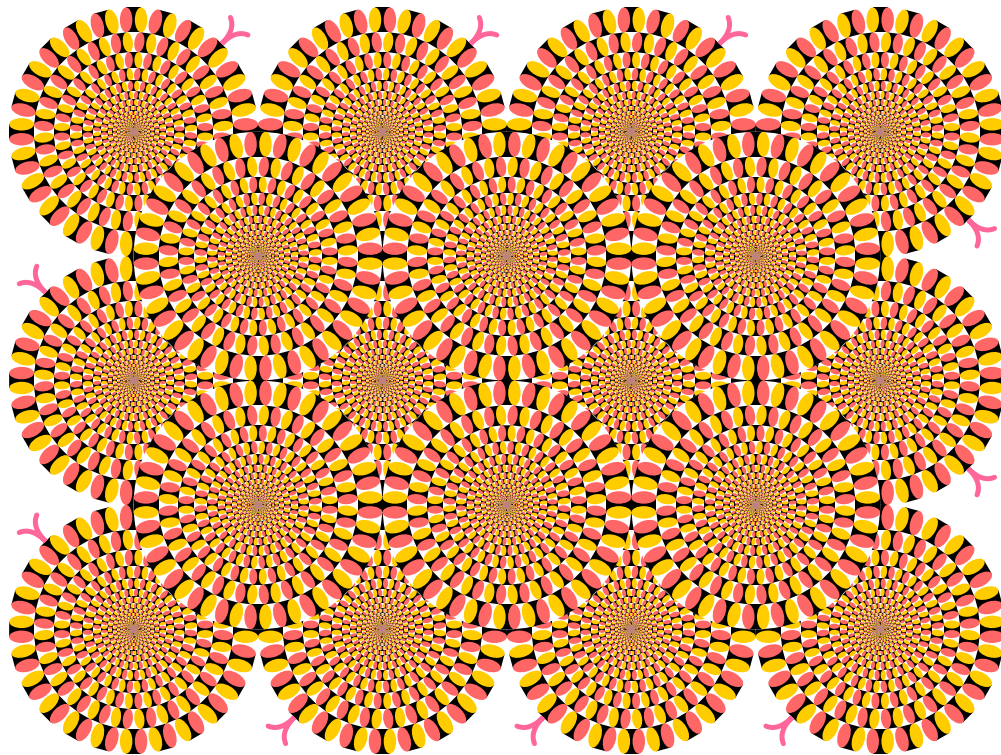
Brighter, A or B? (Checker shadow illusion)



Largest “orange” (Ebbinghaus illusion)



Is It Moving?



Replicating the Vision Processing Is Not Trivial

- Humans process what they sense (perception)
- Computers don't assume (they just take values)
- Seeing is not exactly the same as measuring intensities
- Seeing is building a perception (interpreting)
- In the early days, cameras were further away from human vision
- Today, still far away but not so far (more processing)

More Difficulties

- Occlusion
- Clutter
- Scale
- Deformation
- Lots of classes
- Object's pose
- Computationally expensive

Photos in The Old Days (1880/1895)



Source: www.mcnny.org

Selfies in The Old Days (1920)



Photos Today



Evolution of The Camera

A MINIMALIST GUIDE TO THE EVOLUTION OF THE CAMERA



1814s
Camera Obscura



1837
Daguerreotype
Camera



1888
First Camera
with Film



Circa 1935
35mm Film
Camera



1930s
SLR Cameras



1948
First Polaroid
Model



1980s
35mm Point &
Shoot Cameras



1990s
Digital
Compacts



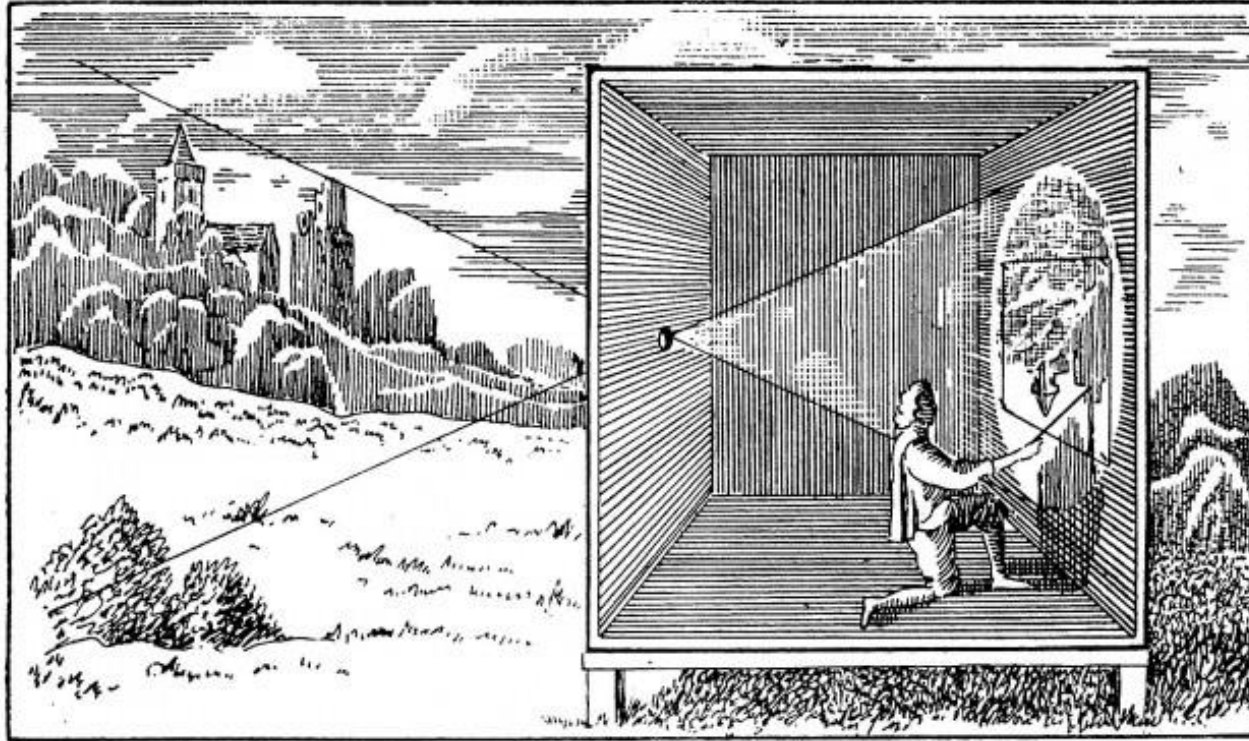
Circa 2007
Slimline
Compacts



Present
Smart Phones

Source: www.opticscentral.com.au

Pinhole Camera (I. E. Camera Obscura)



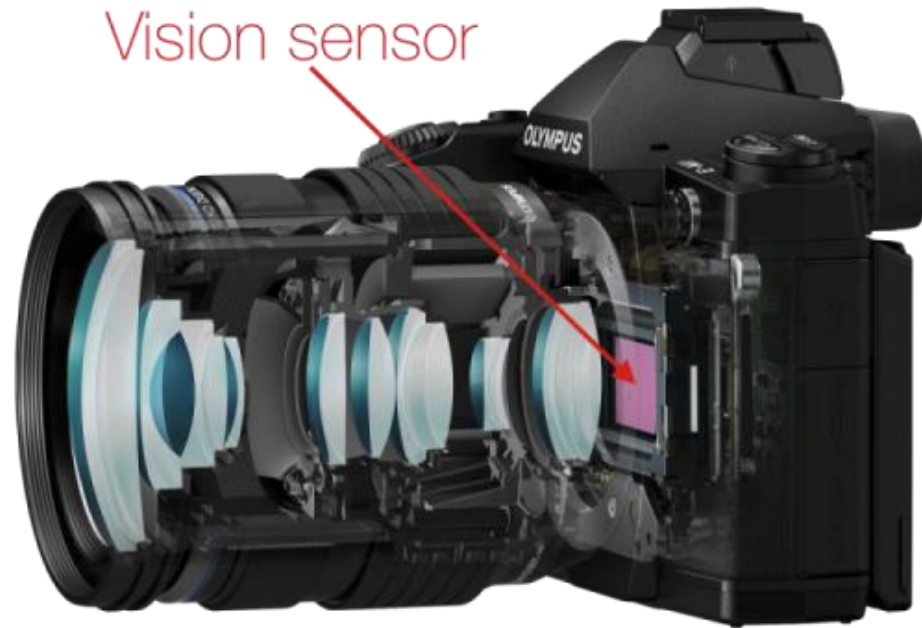
Source: <http://pixslylated.com/blog/stepping-inside-havana-camera-obscura/>

“Pinhole” Apartment

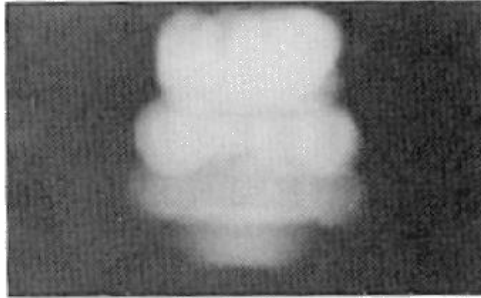


Source: <http://www.abelardomorell.net/project/camera-obscura/>

Camera



Effect of The Aperture



2 mm



1 mm



0.6mm



0.35 mm

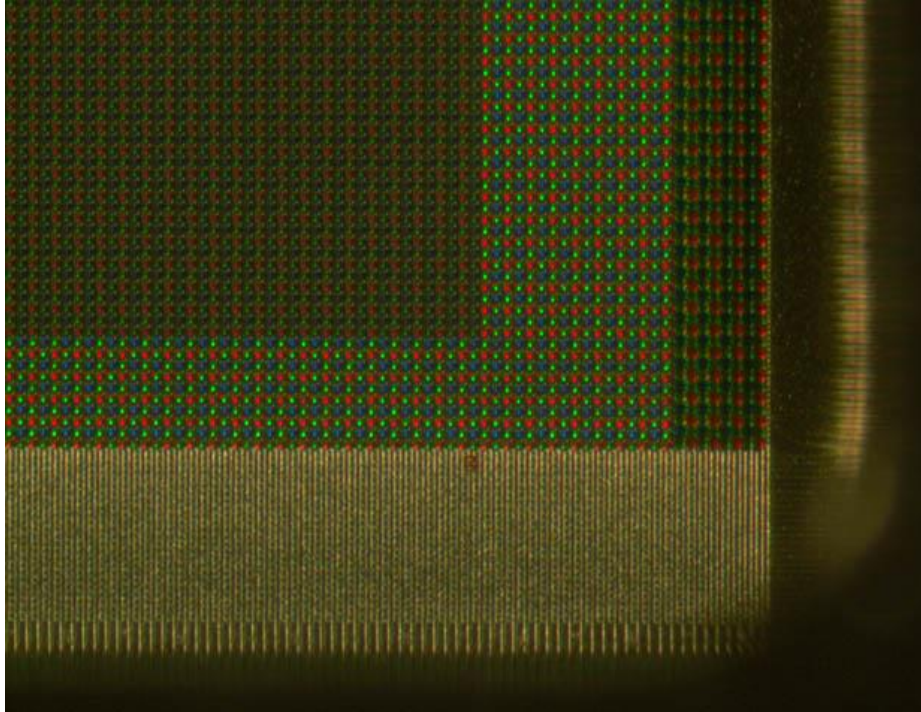


0.15 mm



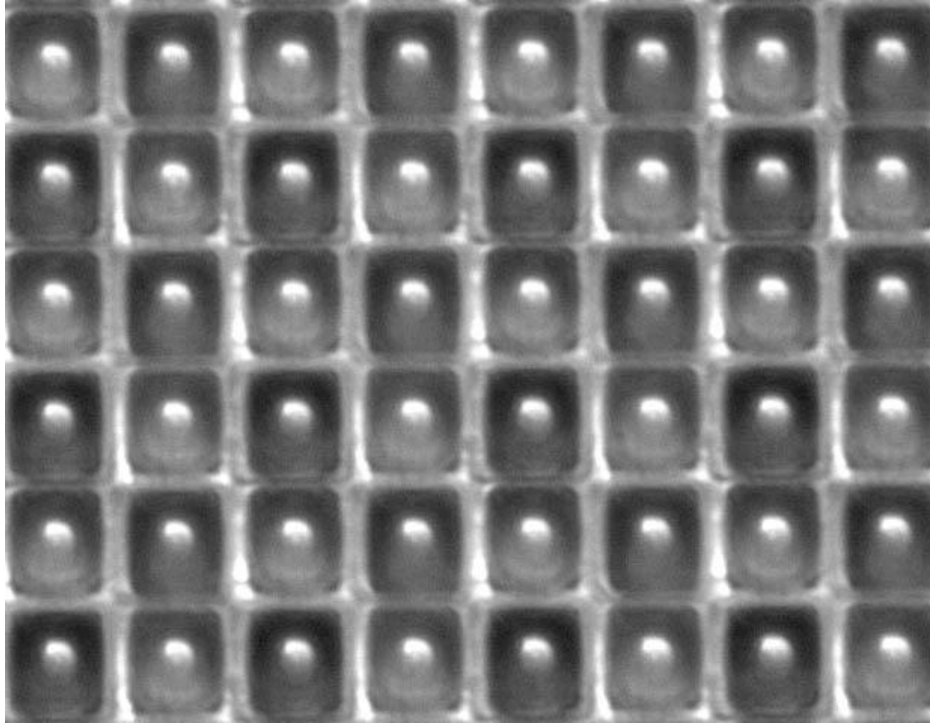
0.07 mm

Back To The Sensor



Source: <https://petapixel.com>

More Zoom



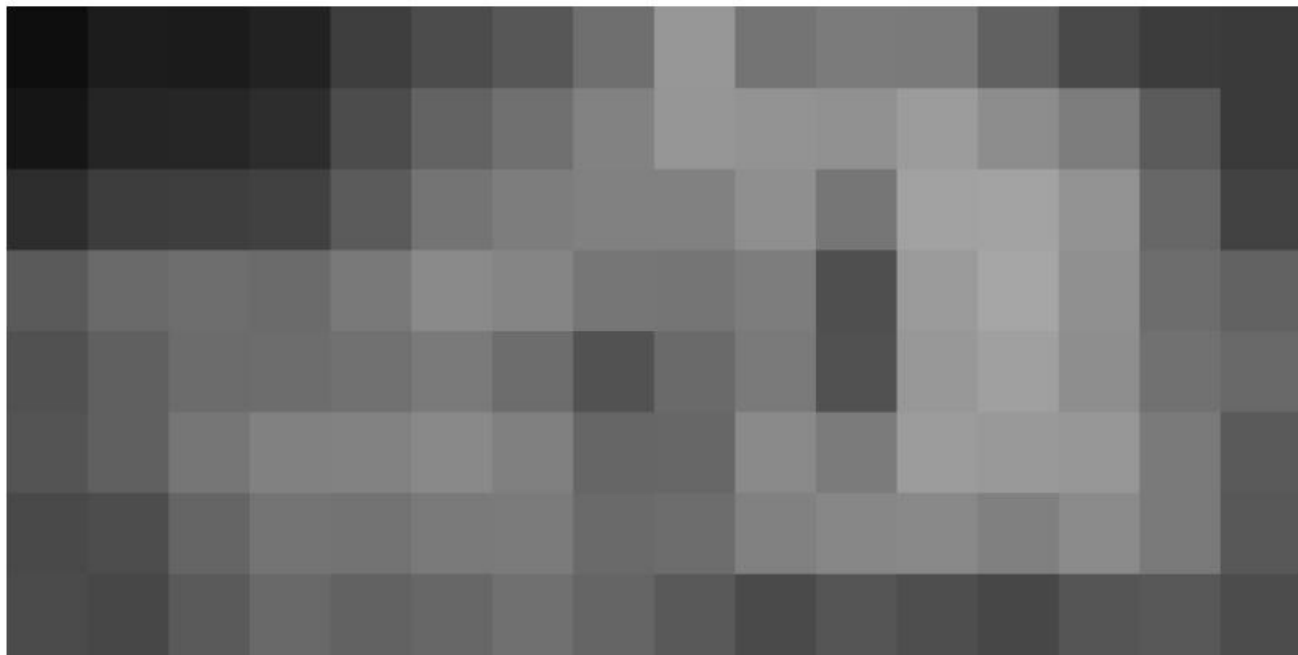
Source: <https://petapixel.com>

Discretization

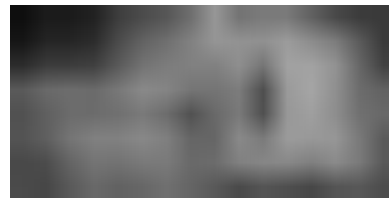
Going from continuous to discrete, involves:

- Sampling (how many bins)
- Quantization (how many levels per bin)

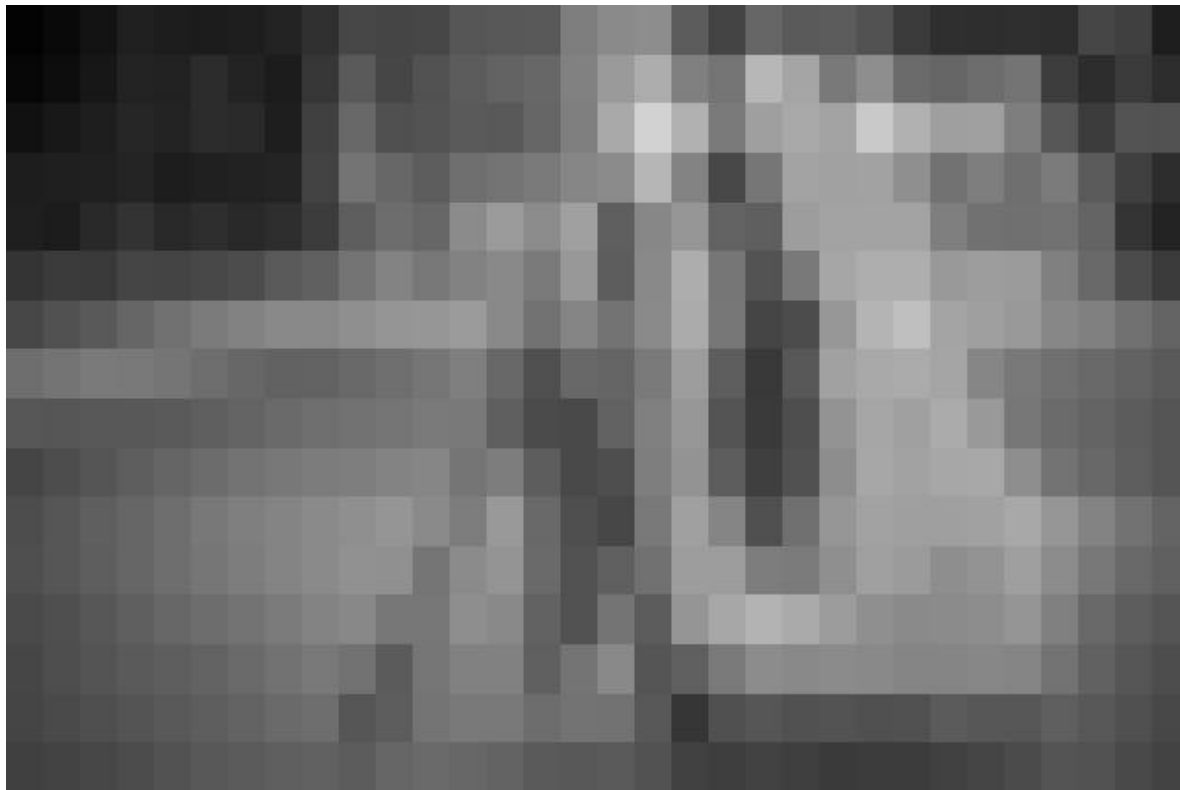
Sampling/Sensor with 128 Bins



Real:



Sensor with 512 Bins



Sensor with 8192 Bins

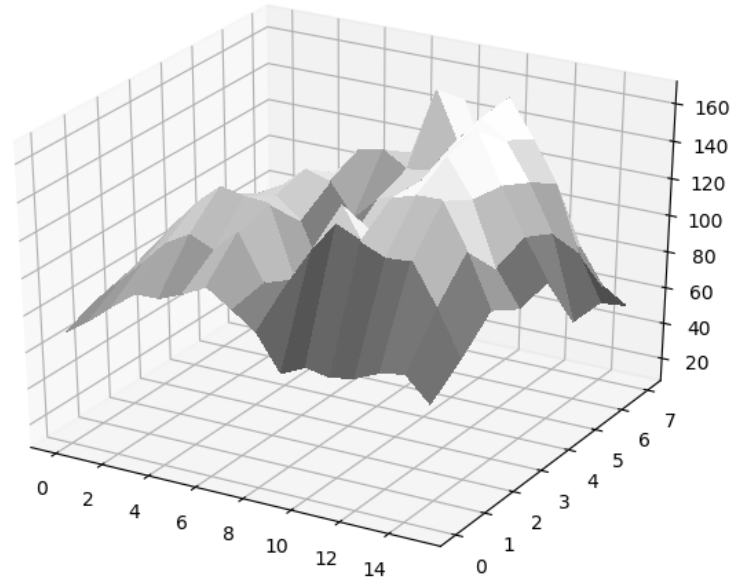


Bins VS Pixels VS Resolution

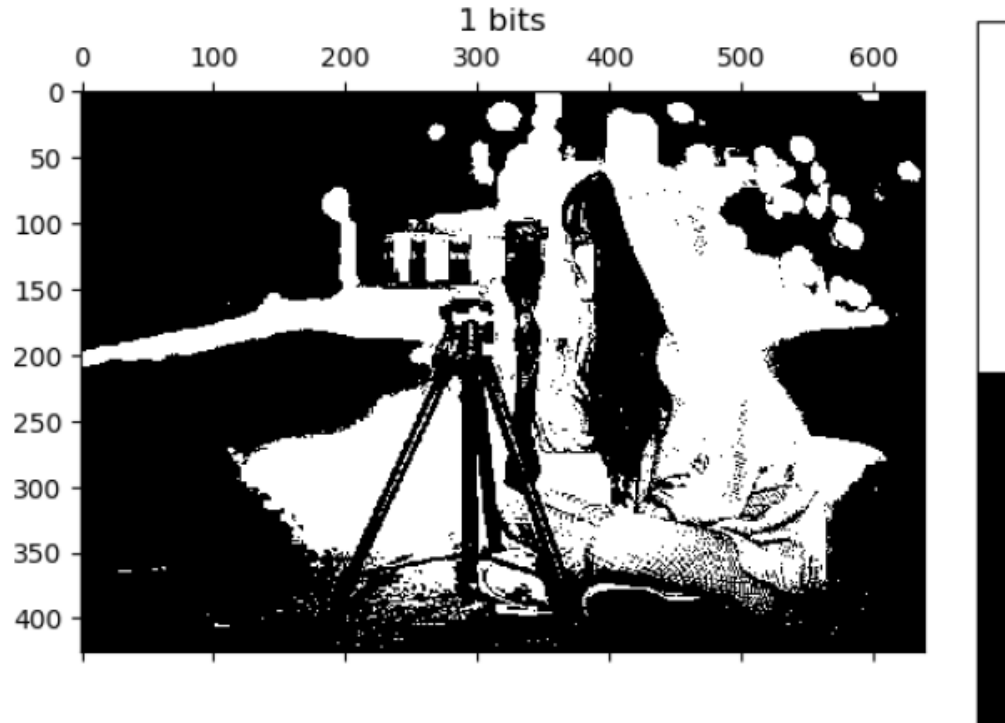
- In general, 1 sensor bin “corresponds” to 1 pixel (picture element), i.e. one reading (or sample)
- Image resolution is the size of the image in pixels
- Image resolutions are, often, expressed as:
 - width x height (pixels)
 - rows x columns (pixels)
 - total pixels (e.g. 12MP)
- When not enough samples are used we start to see artefacts

Images Are Matrices of Numbers

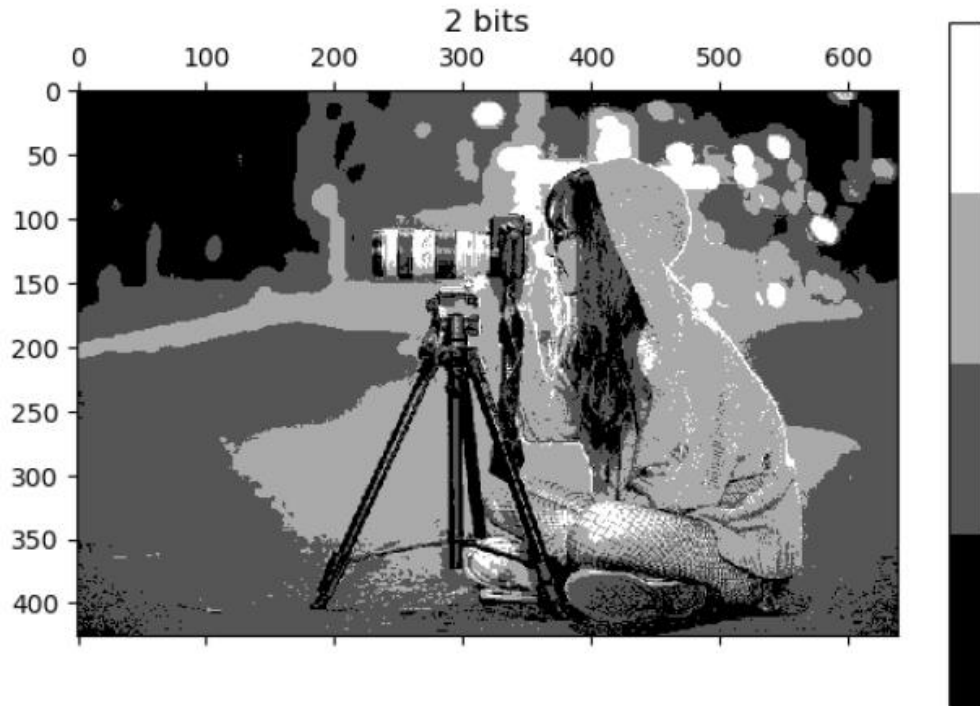
011	027	034	035	064	076	090	114	147	115	128	114	089	080	060	05
025	034	037	040	081	093	104	125	161	133	146	164	145	124	090	06
046	056	061	069	096	118	134	136	128	130	118	162	154	134	110	06
090	101	108	112	119	130	126	109	121	127	087	157	168	142	118	09
084	093	102	111	119	125	115	086	108	117	083	149	164	145	112	09
081	096	112	126	138	135	134	097	104	141	120	150	154	154	131	10
077	089	102	115	122	116	128	105	107	127	144	138	132	133	112	08
069	078	089	098	099	104	110	102	092	073	079	077	079	085	087	07



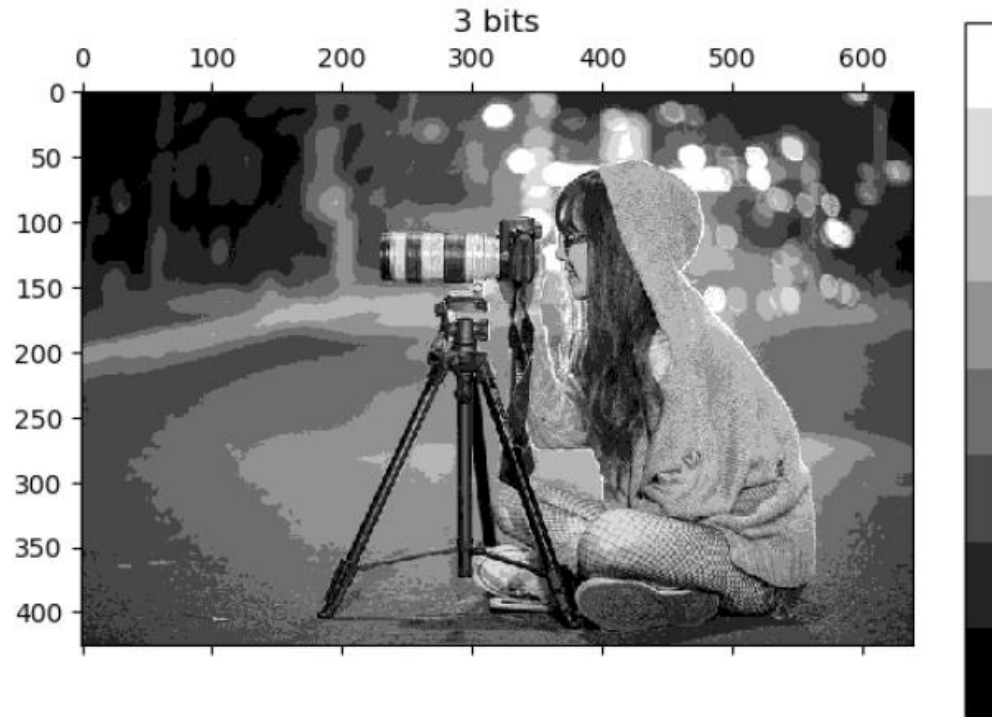
Only 2 Levels (Quantities)



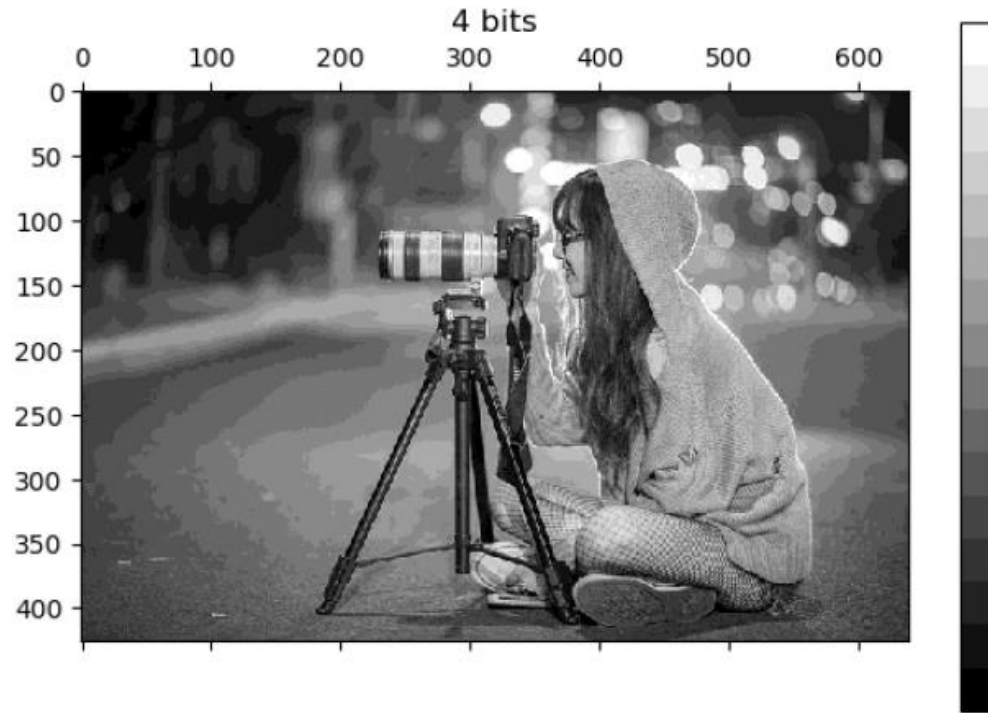
More Levels



More Levels



More Levels



More Levels



More Levels



More Levels



More Levels

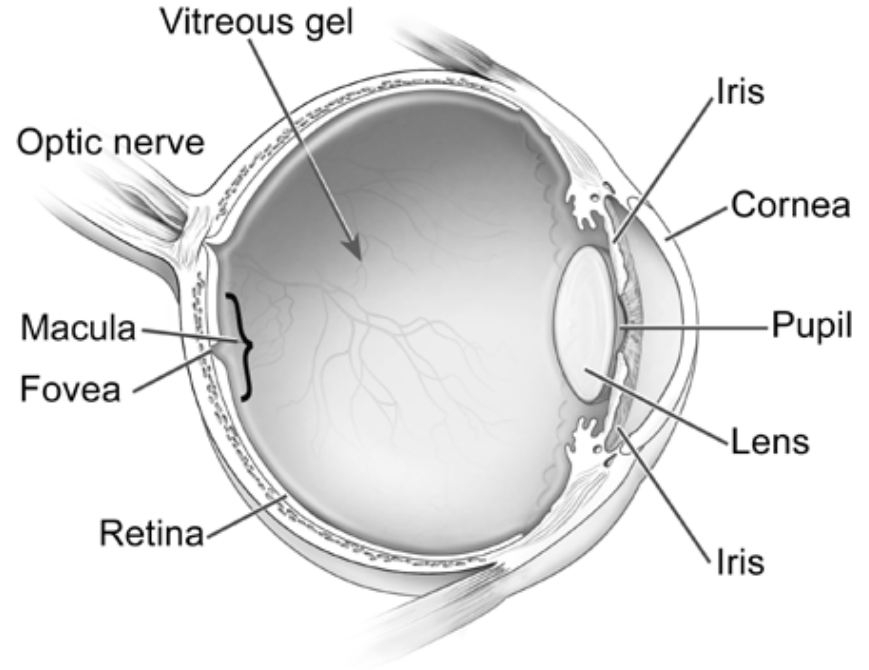


Quantization

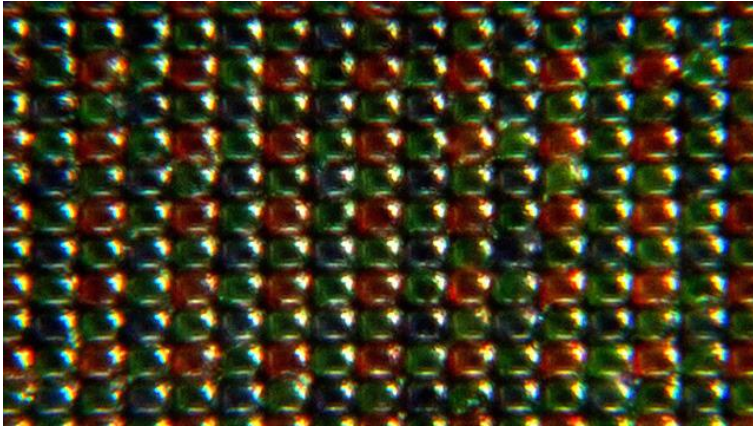
- The number of bits used, is called bit-depth
- Most common bit-depth for grayscale is 8 bits (i.e. 1 byte, unsigned)
- For binary images, it's common to use Boolean
- Quick calculation
 - Number of levels: 2^{bits}
 - Highest intensity: $2^{\text{bits}} - 1$

What Is Color?

- What our visual system perceives as a result of the interaction of light and objects' properties
- Cones are the sensors responsible for color



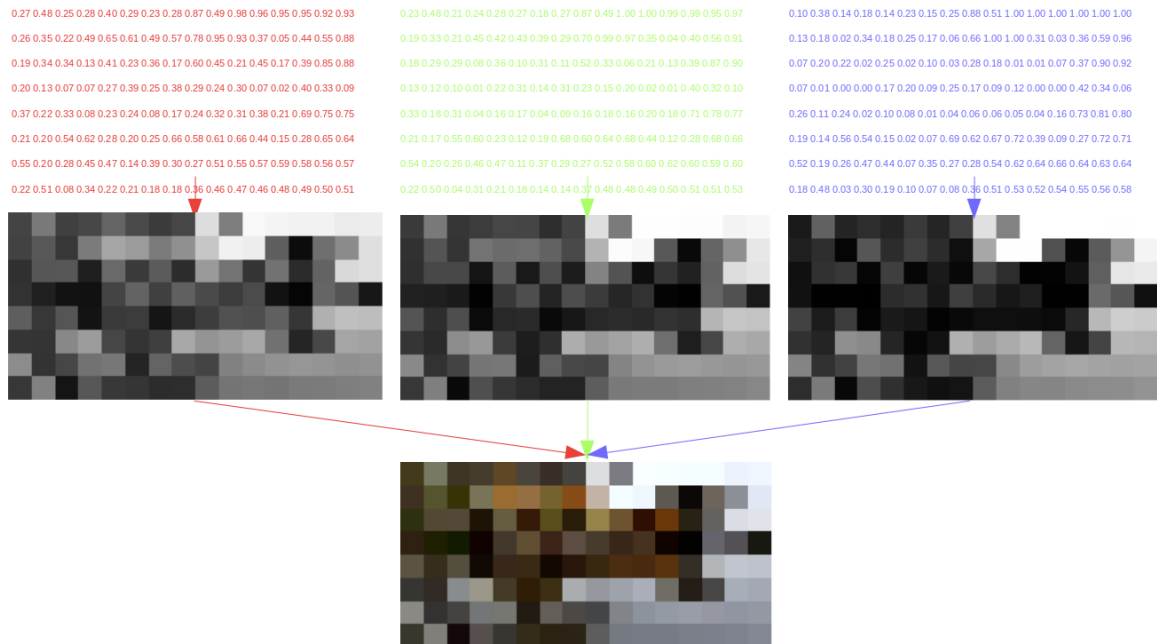
Color Sensors



- Camera sensors use the same principle
 - Per sample ≈ 3 different bins
 - Each bin senses a different wavelength (i.g. R, G, and B)
- Each pixel will have 3 values/readings
 - The color of a pixel consists of linearly combining those 3 values

Color Image

In practice, we will have 3 matrices (AKA channels):



Color to Grayscale

- Based in the human perception of brightness
 - $Y = 0.299R + 0.587G + 0.144B$ (PAL)
 - $Y = 0.2126R + 0.7152G + 0.0593B$ (HDTV, cv2)
 - $Y = 0.2627R + 0.6780G + 0.0583B$ (HDR)
- Alternatives:
 - $Y = (R + G + B)/3$
 - $Y = G$

Summary

- Human vision is difficult to mimic
- Humans give a lot of importance to edges
- The pinhole camera is a very simple model of the eye
- The aperture defines the amount of light hitting the sensor
- The camera sensor samples the scene
- More sensing elements => Higher image resolution
- Sampling-rate is related to how many readings are taken spatially
- The sensor readings must be quantized to be stored as a digital image
- Bit-depth refers to the number of bits used in quantization
- Color is made from a linear combination of 3 Values (RGB)